# Encoding Pen Dynamics in Handwriting Images: An Incremental Study of Parkinson's Disease Classification

Anonymous Authors
*Anonymous Institution*

Keywords: Handwriting, Pseudodynamic Images, Local Phase Quantization, Parkinson's Disease, Explainable AI.

Abstract: We introduce a new formulation of pseudodynamic handwriting minimap signatures, combining contact-only trajectories with normalized pen signals in explicit colour channels. On 597 PaHaW recordings from 75 participants across eight tasks, we compare static traces, four isolated signals (speed, pressure, altitude and azimuth) and four RGB combinations using Local Phase Quantization (LPQ) and SVM, with repeated nested model fitting and common participant partitions. Speed, altitude and azimuth raise observed macro F1 on four tasks, including the spiral (0.5915 to 0.6716), and the task mean (0.5643 to 0.5752). Gains coexist with losses on other tasks; overall superiority remains unresolved after multiplicity correction. In an exploratory comparison on this SVM-selected encoding, LPQ-based classifiers outperform a small CNN trained from scratch; SVM reaches 0.5752 versus 0.5105, with Holm-adjusted $p = 0.0054$. Majority voting yields participant-level macro F1 of 0.6498 for logistic regression and 0.6488 for SVM, a distinct endpoint from the task mean. Held-out Grad-CAM concentrates on writing, yet its median correlation with occlusion is $-0.0121$. The proposed technique provides a reproducible way to examine spatially encoded pen signals, with observed gains that vary across tasks and classifier choices.

## 1 INTRODUCTION

A digitizing tablet records the pen trajectory together with changes in speed, pressure and inclination. Encoding these measurements as colour adds information about the production of a trace while preserving its recognizable structure. We investigate whether these spatially encoded signals improve classification beyond the visible trajectory.

The application is the distinction between Parkinson's disease (PD) and healthy controls (H) in PaHaW, a database of online handwriting and drawing tasks (Drotár et al., 2016). Handwriting exposes aspects of fine motor behaviour, but the relation between pen motion and diagnosis also depends on the task, device and participant characteristics (Impedovo and Pirlo, 2019).

We propose a new formulation of *pseudodynamic handwriting minimap signatures*: compact visual representations of task executions, derived from dynamic recordings. The term minimap follows work using compact source-code renderings as visual signatures (Gebara Jr. et al., 2025). Here, planar position supplies the image coordinates and colour carries measured pen attributes. A signature denotes this representation, not a person's authentication signature. Starting from a monochrome trace, we add speed, pressure, altitude and azimuth separately, then compare all four three-signal RGB combinations. This progression makes the contribution of each encoding an explicit experimental question.

Handwriting classification has often used explicitly designed kinematic and pressure features. Drotár et al. compared such features with several classifiers on PaHaW (Drotár et al., 2016). Learned features provide another route, either from pen dynamics (Pereira et al., 2016) or from visual attributes of the reconstructed writing (Moetesum et al., 2019). These approaches motivate comparing a fixed image descriptor with a network that learns from pixels.

Dynamic information has appeared in earlier handwriting images. Diaz et al. encode movement through acquired points and pen-up trajectories for PD detection (Diaz et al., 2019). Cilia et al. construct synthetic RGB images from handwriting dynamics for Alzheimer's disease detection, comparing them with binary images using deep transfer learning (Cilia et al., 2021). These precedents establish the broader idea of expressing dynamics visually.

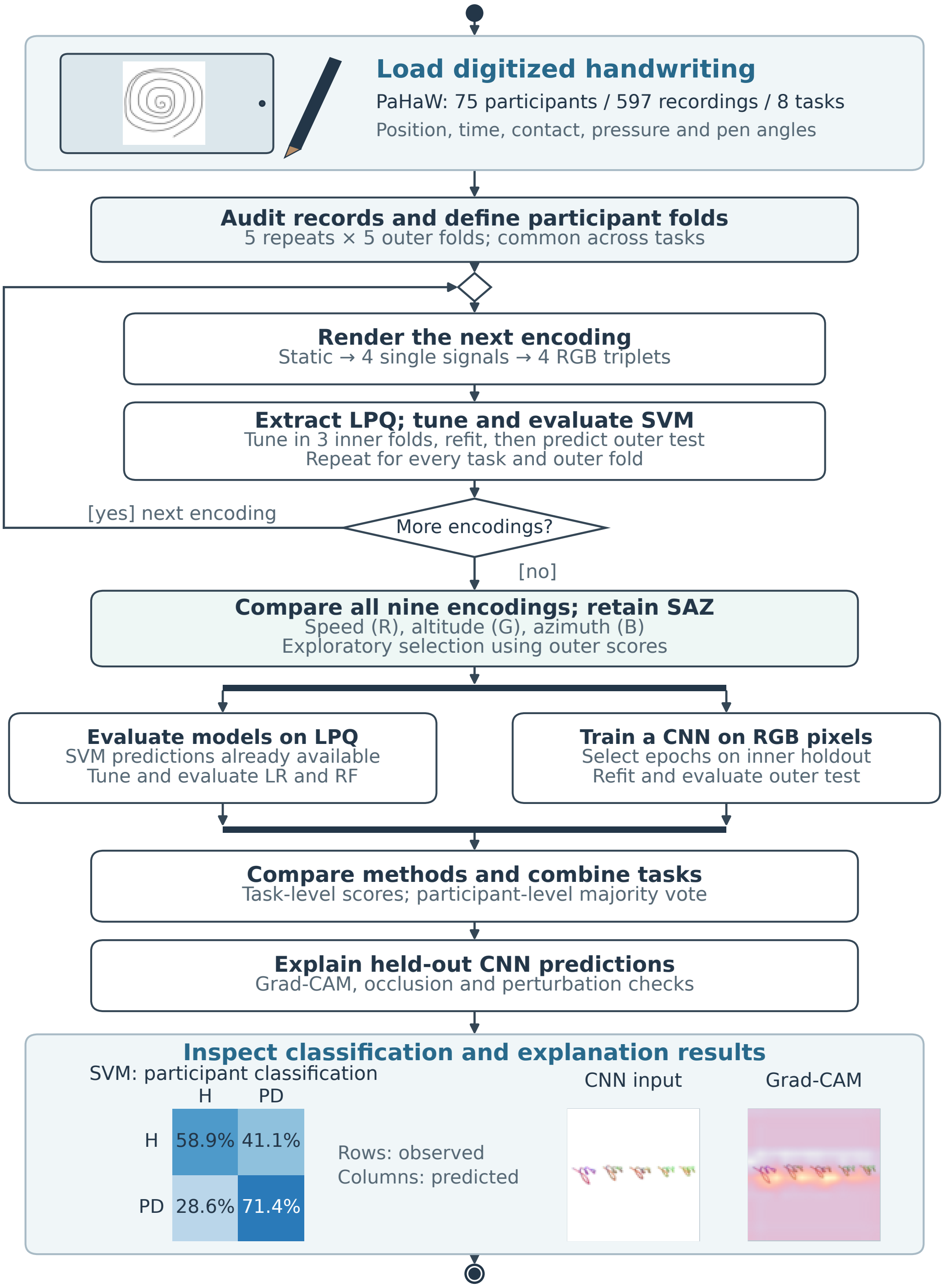


Figure 1: Illustrated activity diagram of the incremental study. Nine encodings are evaluated with LPQ and SVM using participant-separated outer folds. Selecting *SAZ* from the outer scores is exploratory. The classifier branches reuse the participant assignments; LR denotes logistic regression and RF random forest. Task scores and participant voting are distinct endpoints. XAI uses the CNN checkpoints responsible for the held-out predictions. The final illustrations show an aggregate SVM confusion matrix and one previously selected CNN explanation, from different analyses. The tablet and stylus are schematic; the handwriting and results are from the recorded experiments.

Our formulation combines contact-only minimap rendering, within-recording percentile normalization and circular azimuth centring to map pen signals onto a fixed trajectory. We evaluate its components incrementally across all eight PaHaW tasks, using the same Local Phase Quantization (LPQ) descriptor, support vector machine (SVM) search and participant partitions for nine encodings. All recordings and renderings from a participant remain in the same outer partition; model hyperparameters are selected within the training data. Subsequent analyses examine classifier choice, task fusion and CNN explanations.

The speed–altitude–azimuth encoding raises observed macro F1 on four tasks, with its largest increase on the spiral, from 0.5915 to 0.6716. The task mean rises from 0.5643 to 0.5752, although losses on the other four tasks and the paired uncertainty leave overall superiority unresolved after multiplicity correction.

In the exploratory comparison on this SVM-selected encoding, LPQ with SVM reaches task-mean macro F1 of 0.5752, compared with 0.5105 for a small CNN trained from scratch; the paired difference survives Holm correction. Logistic regression and Random Forest also exceed this CNN.

Majority voting across tasks yields participant-level macro F1 near 0.65 for logistic regression and SVM. This endpoint describes one decision per person. For the CNN, held-out Grad-CAM maps concentrate on writing and deleting high-attribution pixels changes prediction confidence, yet their median spatial correlation with occlusion is close to zero.

Representation selection and subsequent comparisons use the same cohort; the evaluation does not independently validate the complete selection procedure.

Figure 1 shows the encoding loop and the subsequent classifier and XAI analyses under the common participant partitions.

# 2 BUILDING THE REPRESENTATION

To vary the information encoded in each signature, we first reconstruct the visible trace. We then map individual pen signals to colour and combine them without changing the trajectory or rendering geometry.

## 2.1 The Visible Trace

Let a recording contain samples

$$q_i = (x_i, y_i, t_i, s_i, z_i, a_i, p_i), \tag{1}$$

where $s_i$ indicates surface contact, $z_i$ is azimuth, $a_i$ is altitude and $p_i$ is pressure. The stored PaHaW columns place $y$ before $x$; the renderer restores their planar order.

The static image uses only position and the contact indicator. A segment between samples $i$ and $i+1$ is drawn if both samples are on the surface:

$$\mathcal{S} = \{i : s_i = 1 \text{ and } s_{i+1} = 1\}. \tag{2}$$

This preserves pen lifts as gaps instead of joining the end of one stroke to the beginning of the next. Neither the in-air trajectory nor the duration of a lift appears in the image. Contact points alone determine its bounding box, so an in-air excursion cannot change the scale of the visible writing.

The bounding box is centred on a $512 \times 512$ white canvas with a 20-pixel margin, using one scale factor for both axes. Segments are drawn in RGB $(30, 30, 30)$ with constant 4-pixel thickness. Rendering takes place at three times the target resolution and is reduced with Lanczos filtering. These settings remain fixed throughout the colour experiments.

The image retains form and stroke discontinuities, but fitting every record to its own canvas removes absolute size. Geometrically similar samples of different sizes can produce the same image, discarding information about absolute micrographia.

## 2.2 Adding One Signal

Four measurements are candidates for colour: speed ($S$), pressure ($P$), altitude ($A$) and azimuth ($Z$). Pressure is the tablet's contact-pressure signal. Altitude describes pen inclination relative to the surface, while azimuth describes its orientation in the tablet plane. We first encode each signal alone, always in the red channel, with green and blue fixed at 30. This avoids changing the channel assignment while screening the signals.

For a surface segment, speed is calculated directly from the consecutive coordinates and timestamps:

$$v_i = \frac{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}}{t_{i+1} - t_i}. \tag{3}$$

We use $\log(1 + v_i)$ in the tablet's native coordinate and time units. This compresses large values before the intensity mapping. Pressure and altitude use the arithmetic mean of the two endpoint measurements.

Azimuth requires a circular reference. We convert the recorded tenths of a degree to radians and compute the mean direction over contact samples. Each angle is centred on that direction and wrapped into $[-\pi, \pi]$ before the endpoint average is taken. This reduces the effect of an arbitrary zero-degree boundary. It is a

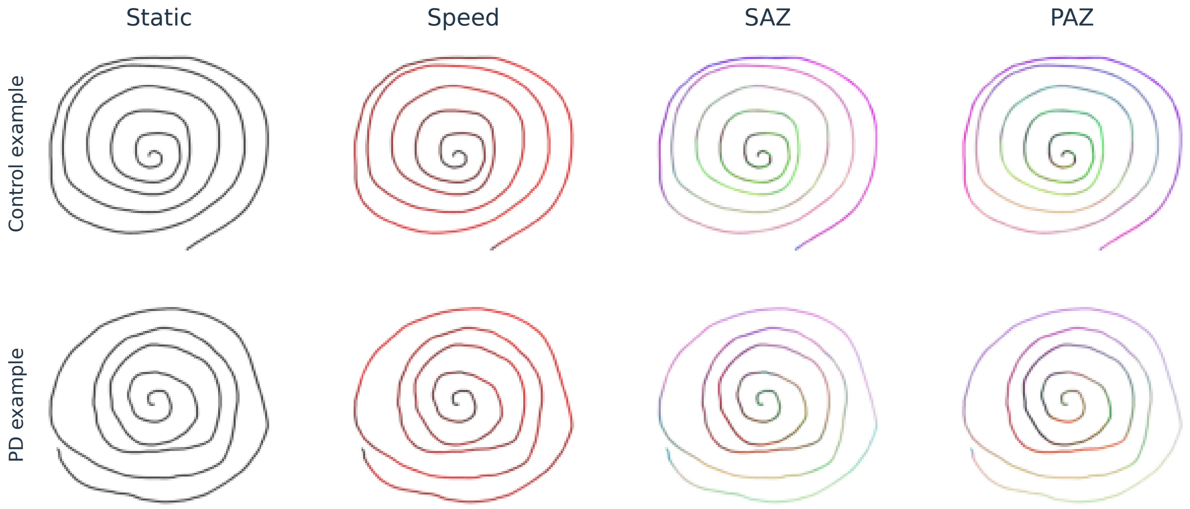


Figure 2: Building a pseudodynamic image from the same spiral trace. Rows show one control and one PD example, selected by a fixed identifier rule without reference to prediction quality. Columns move from static geometry to speed alone and two RGB triplets. The trajectory is unchanged; colour describes within-recording variation.

circular centring operation, not temporal unwrapping, and it does not remove every possible branch-cut artefact.

Each transformed signal is scaled within the recording being rendered. Let $q_{.05}$ and $q_{.95}$ be its fifth and ninety-fifth percentiles over surface segments. The normalized value is

$$u_i = \text{clip}\left(\frac{f_i - q_{.05}}{q_{.95} - q_{.05}}, 0, 1\right), \tag{4}$$

with $u_i = 0.5$ if the percentile range is degenerate. We map it to an integer channel value

$$c_i = \text{round}(30 + 190u_i). \tag{5}$$

The white background stays outside the semantic range. Antialiasing introduces intermediate intensities at boundaries, as in the static reconstruction.

Each record is normalized independently, without a cohort-wide reference distribution or class labels. Colour therefore describes relative variation within a recording: the same red value in two images need not represent the same absolute speed or pressure. Circular centring also removes absolute pen orientation. These quantities remain available to methods that use the physical measurements directly.

## 2.3 Combining Signals in RGB

We assign one scalar signal to each RGB channel and evaluate all four subsets of size three. Table 1 gives the assignments, and Figure 2 illustrates the progression. All combinations use the same geometry, background and rendering settings as the static and isolated-signal images.

Table 1: Incremental colour encodings. A dash denotes a constant channel value of 30. $S$: speed; $P$: pressure; $A$: altitude; $Z$: azimuth.

| Encoding | Red | Green | Blue |
|---|---|---|---|
| Static | – | – | – |
| One signal | $S$, $P$, $A$ or $Z$ | – | – |
| *SPA* | $S$ | $P$ | $A$ |
| *SPZ* | $S$ | $P$ | $Z$ |
| *SAZ* | $S$ | $A$ | $Z$ |
| *PAZ* | $P$ | $A$ | $Z$ |

The comparison holds the classifier procedure fixed. Because its descriptor concatenates the same histogram construction for each channel, permuting the RGB channels only permutes three feature blocks. Standardization followed by a linear or radial-basis SVM is invariant to that common permutation, apart from numerical effects. The RGB assignments thus specify a reproducible representation without requiring six permutations of every triplet for the SVM experiment. This argument does not extend to every image model; the later CNN receives one fixed RGB assignment.

Segments are drawn in acquisition order. A later segment can overwrite an earlier colour at a crossing, and the image has no separate layer for repeated visits to the same position. It preserves spatially located signal variation, not the complete sample sequence. Calling the result pseudodynamic reflects that distinction.

Table 2: Task inventory. T1 contains 36 PD and 36 H records; each other task contains 37 PD and 38 H records.

| ID | Task | Records |
|---|---|---|
| T1 | Archimedean spiral | 72 |
| T2 | Repeated *l* | 75 |
| T3 | Repeated *le* | 75 |
| T4 | Repeated *les* | 75 |
| T5 | Repeated *lektorka* | 75 |
| T6 | Repeated *porovnat* | 75 |
| T7 | Repeated *nepopadnout* | 75 |
| T8 | *Tramvaj dnes už nepojede.* | 75 |

# 3 DATA AND EVALUATION

Each recording produces nine alternative images, all derived from the same participant. We compare encodings on common participant partitions and define separate endpoints for individual tasks and decisions combined across tasks.

## 3.1 Participants and Tasks

PaHaW provides eight tasks recorded with a Wacom Intuos 4M tablet (Drotár et al., 2016). We calculate speed from the recorded timestamps rather than an assumed sampling interval. Our local inventory contains 597 recordings from 75 participants: 37 with PD and 38 controls. Table 2 specifies the task set. Three spiral recordings are absent from the local corpus; the other tasks have one recording for every participant. We retain all available records and do not interpret missing files as quality exclusions.

An input audit checked file counts, sample counts, finite values and timestamp increments. Five files have a declared sample count different from the number of rows present. The renderer uses the rows actually available. There are 253 negative timestamp increments in 151 recordings, but all 1,065,788 segments with contact at both endpoints have positive increments. Equation 3 therefore requires no interpolation or imputation for the segments used in this experiment. None of the signal normalization ranges was degenerate.

The participant is the independent observational unit; tasks and alternative renderings do not enlarge the cohort. Mean age is 69.32 years in PD and 62.42 in H. Age and clinical fields are excluded from classifier inputs, but handwriting may still carry this confounding.

## 3.2 A Common Partition for Every Task

We construct stratified folds from one row per participant, then propagate the assignments to every task and rendering (Figure 3). Five repetitions of five outer folds produce a test prediction for every available record in each repetition. All tasks from a participant are in the same outer fold. The spiral task uses the participants for whom a record exists.

Within each outer training partition, three participant-separated inner folds choose the classical model hyperparameters. Feature standardization is fitted on the corresponding training portion, never on its validation or test participants. The selected model is refitted on all outer training participants and evaluated once on the outer test set. The rendering is independent per record, so it has no parameters learned from the other records before splitting.

Each fixed encoding and classifier yields 200 outer models and 2,985 outer-test predictions. These are repeated predictions of 597 records from 75 participants. Every later stage reuses the saved assignments.

The highest-ranked encoding was selected from these outer scores for subsequent comparisons. Nested fitting protects model hyperparameter estimation, but not that representation choice. Earlier explorations also used PaHaW. The downstream results thus concern a fixed, selected encoding on a development cohort, without an independent estimate of the complete selection procedure.

## 3.3 Metrics and Paired Comparisons

Macro F1 gives equal weight to the two diagnosis classes:

$$F_{\text{macro}} = \frac{F_{1,\text{H}} + F_{1,\text{PD}}}{2}. \tag{6}$$

For each task and repetition, we concatenate the five outer-test folds and compute the metric once. We then average the five resulting scores. For the representation and classifier rankings, the endpoint is the unweighted mean across all eight tasks:

$$\overline{F}_{\text{tasks}} = \frac{1}{8 \cdot 5} \sum_{t=1}^{8} \sum_{r=1}^{5} F_{t,r}. \tag{7}$$

We also retain accuracy, balanced accuracy, PD sensitivity, H specificity, positive-class F1 and ROC AUC. Section 6 defines a separate participant-level prediction by combining tasks.

We calculate 95% percentile confidence intervals from 10,000 diagnosis-stratified bootstrap draws of participants. A draw preserves all available tasks and repeated predictions for each selected person. For a paired contrast, the same draw is applied to both methods. The randomization test exchanges method predictions per participant, jointly across tasks and repetitions, using 10,000 participant-wise swap patterns. It tests the difference between fixed sets of predictions, not a shuffled-label learning pipeline.

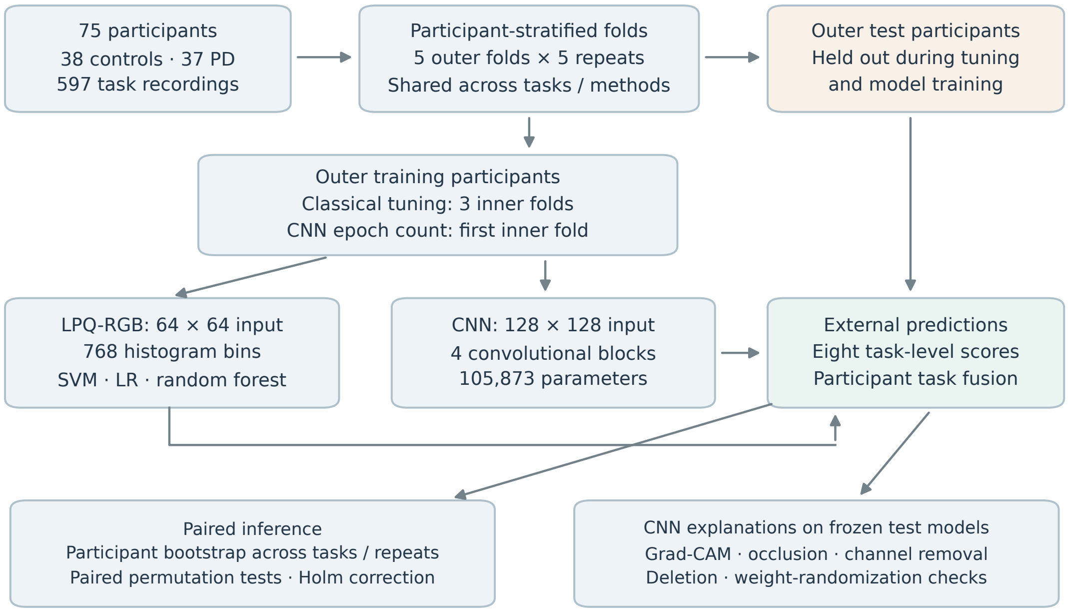


Figure 3: Participant separation throughout model fitting and evaluation. The CNN selects its epoch count on the first inner holdout; the classical models average three inner-fold scores. The outer assignments are common across tasks and methods. This diagram describes model fitting for a fixed encoding; representation selection from the outer scores remains exploratory.

Holm adjustment covers 36 encoding pairs; two separate 32-contrast task families for isolated signals and triplets against static; six classifier pairs at each endpoint; three SVM fusion-rule pairs; and eight task-specific CNN–SVM contrasts. Correction is within each declared family, not across all decisions in the broader research history. The intervals are pointwise, not simultaneous.

Both resampling procedures leave models fixed. Intervals and tests describe conditional uncertainty in these predictions, without the full variability of re-training or selection.

# 4 FROM SHAPE TO COLOUR

The common participant partitions allow paired comparisons from static traces to isolated signals and RGB combinations. We hold LPQ extraction and SVM selection fixed so that each comparison changes the image encoding.

## 4.1 A Fixed Descriptor for the Comparison

Local Phase Quantization describes texture through local Fourier phase information (Ojansivu and Heikkilä, 2008). We apply the same implementation independently to each RGB channel.

Before extraction, foreground is detected by any channel intensity below 250. We crop its bounding box, add a margin equal to 5% of the longer dimension, pad to a square with white, and resize to $64 \times 64$ by bilinear interpolation. LPQ filters each channel with four complex $5 \times 5$ kernels at frequency offsets $(1,0)$, $(0,1)$, $(1,1)$ and $(1,-1)$, scaled by $1/5$. The signs of their real and imaginary responses form an eight-bit code at each pixel, using reflected boundaries. We normalize a 256-bin histogram by its count, take its element-wise square root, and concatenate the channels. The 768-attribute vector includes the padded background. No decorrelation or spatial pyramid is applied.

The static RGB channels are identical, so the static vector repeats the same histogram three times. This keeps the interface and nominal dimension constant; it does not create additional static information. For the isolated signals, two channels remain constant along the trace and retain its geometric pattern.

Standardization and SVM fitting occur inside each training partition. The search contains five linear models with $C \in \{0.01, 0.1, 1, 10, 100\}$ and sixteen radial-basis models with $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{\text{scale}, 10^{-4}, 10^{-3}, 10^{-2}\}$. We use inverse-frequency class weights. The candidate with the highest mean inner-fold macro F1 is chosen, with ties resolved by the stored candidate order. The SVM learns from image texture; it never receives the original signal measurements as tabular features (Cortes and Vapnik, 1995).

## 4.2 The Static Baseline

The static reconstruction reaches a task-mean macro F1 of 0.5643. Its best task is repeated *les* (T4), with 0.6502 and a participant-bootstrap interval of [0.5582, 0.7351]. The spiral reaches 0.5915 and T5 reaches 0.5038. All eight tasks remain in the subsequent encoding comparison.

## 4.3 Isolated Signals and RGB Combinations

Figure 4 orders the encodings by Equation 7. Altitude gives the highest task mean among the isolated signals at 0.5718, followed by azimuth at 0.5688. Speed alone reaches 0.5625 and pressure 0.5568, both below the static mean of 0.5643.

The individual-task contrasts vary in direction. For example, speed changes the spiral score from 0.5915 to 0.6153 but changes repeated *les* from 0.6502 to 0.5994. Altitude increases the spiral score to 0.6294 and changes T4 only slightly, to 0.6548. None of the 32 isolated-signal versus static contrasts survives its Holm correction.

Combining signals produces a similarly narrow ranking. Starting from *PAZ*, replacing pressure, azimuth or altitude with speed gives *SAZ*, *SPA* or *SPZ*, respectively. *SAZ* leads at 0.5752, only 0.0006 above *PAZ*; *SPA* reaches 0.5697 and *SPZ* reaches 0.5574. These results provide no basis for declaring pressure intrinsically uninformative. LPQ summarizes normalized image patterns, not the physical measurements directly.

The paired *SAZ*–static difference is +0.0109 with a 95% interval of [−0.0146, 0.0370]. Its randomization $p$ is 0.4096 and Holm-adjusted $p$ is 1.0000 across the 36 encoding pairs. The *SAZ*–*PAZ* interval is [−0.0185, 0.0198], also with adjusted $p = 1.0000$. We carry *SAZ* forward as the numerical leader in this exploratory comparison.

Its per-task changes explain the smaller average gain: the spiral improves by 0.0802, T6 by 0.0524, and T7 by 0.0455, whereas T2 falls by 0.0612. None of its eight task-specific contrasts against static remains significant within the 32-comparison triplet family.

# 5 FROM TEXTURE TO LEARNED FEATURES

The encoding comparison leads to a second question: how does the selected representation interact with the classifier? Keeping *SAZ* and the participant partitions fixed, we compare models fitted to LPQ features with a CNN that learns directly from pixels.

## 5.1 Linear and Tree-Based Alternatives

With the *SAZ* image fixed, SVM, regularized logistic regression and Random Forest use the same 768 LPQ attributes. Logistic regression provides a linear baseline. We search $L_1$ and $L_2$ penalties with $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$, using the liblinear solver and inverse-frequency class weights. Standardization remains inside the training pipeline.

Random Forest supplies a different form of nonlinear modelling through an ensemble of decision trees (Breiman, 2001). Each candidate uses 100 trees, the Gini criterion and class weights calculated within bootstrap samples. The search combines maximum-feature choices $\{\sqrt{d}, 0.25d\}$ with minimum leaf sizes $\{1, 2, 4\}$, where $d = 768$. Tree depth is unrestricted. Both algorithms reuse the same three-fold inner search and the same outer test assignments as SVM.

## 5.2 Learning Directly from Pixels

We use a compact CNN trained from scratch as a direct pixel-learning baseline, following the use of convolutional models in document recognition (LeCun et al., 1998). Its architecture is fixed throughout the comparison.

The input follows the same foreground crop and square padding as LPQ but is resized to $128 \times 128$. Pixel values are mapped from $[0, 255]$ to $[-1, 1]$ with fixed constants. Four blocks contain a bias-free $3 \times 3$ convolution, batch normalization, ReLU and $2 \times 2$ max pooling. Their channel counts are 16, 32, 64 and 128. A dropout layer with probability 0.3 and one linear output act on the flattened $128 \times 8 \times 8$ feature map. The network has 105,873 trainable parameters.

Training uses AdamW with learning rate $10^{-3}$, weight decay $10^{-4}$, batch size 16 and binary cross-entropy with positive weight equal to the H/PD training-count ratio. Training augmentation uses rotations up to 3°, translations up to 3% and scale changes between 0.95 and 1.05. White fills exposed pixels; writing is never flipped.

For each outer fold and task, the first of the saved inner splits chooses the number of epochs. Selec-

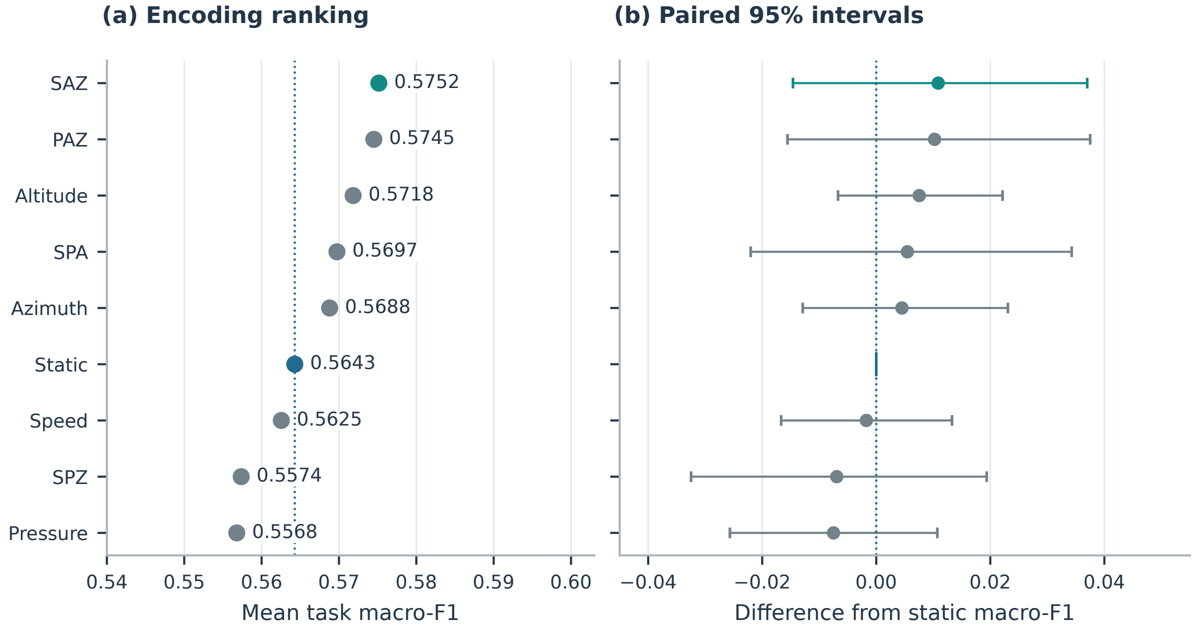


Figure 4: Encoding comparison with SVM. Left: unweighted mean of task-specific macro F1 over eight tasks and five repetitions. Right: paired differences from static with conditional 95% participant-bootstrap intervals. The complete family of 36 pairwise comparisons supports no difference after Holm adjustment.

tion uses validation macro F1, with binary cross-entropy as the tie-breaker. Training runs for at most 60 epochs, with a patience of ten and stopping enabled after epoch twelve. The selected epoch itself may be earlier than twelve. We then reinitialize the network and train on all outer training participants for exactly that number of epochs. The outer test set is used only for the saved evaluation. Unlike the classical grid searches, the CNN epoch selection uses one inner holdout rather than an average over all three inner folds.

Selected counts range from 1 to 48 epochs, with median 15. All 200 outer checkpoints are retained for XAI.

### 5.3 Task-Level Comparison

SVM leads the task-mean ranking at 0.5752, followed by logistic regression at 0.5674 and Random Forest at 0.5599. Their differences are small relative to the paired uncertainty. SVM minus logistic regression is 0.0077, with interval $[-0.0074, 0.0230]$ and Holm-adjusted $p = 0.6563$. SVM minus Random Forest is 0.0152, with interval $[-0.0066, 0.0365]$ and adjusted $p = 0.5645$.

The CNN reaches 0.5105. The SVM–CNN contrast is 0.0646, with interval $[0.0277, 0.1021]$ and adjusted $p = 0.0054$ in the six-pair classifier family. Random Forest and logistic regression also exceed the CNN, with adjusted $p = 0.0460$ and $p = 0.0255$, respectively. The inference follows the fixed-prediction procedure in Section 3.

The task pattern in Figure 5 differs from the overall ranking. Random Forest is strongest on T4 at 0.6791, even though its eight-task mean is below SVM. The CNN is slightly above SVM on T2 and T5, two tasks on which SVM itself performs poorly. Its task-specific differences against SVM do not survive the family of eight contrasts.

LPQ-based models perform better under these choices, which differ in more than architecture: LPQ uses $64 \times 64$ inputs, the CNN uses $128 \times 128$, and their selection budgets differ. The comparison does not isolate the effect of architecture or evaluate pre-trained features such as those used by Diaz et al. (Diaz et al., 2019).

## 6 COMBINING HANDWRITING TASKS

Late fusion combines the available task predictions into one decision per participant. Because all tasks share the same participant folds, every model contributing to a held-out person's decision was fitted without that person.

We first considered three SVM rules over every available task: the mean calibrated PD proba-

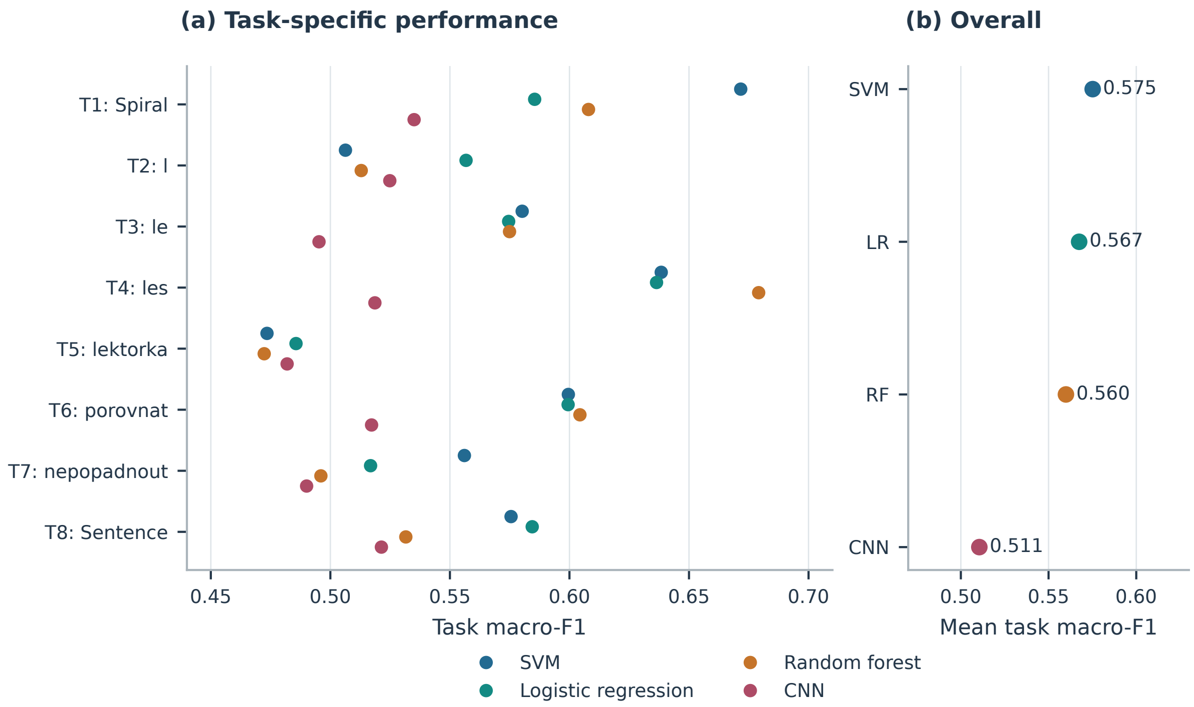


Figure 5: Classifier comparison on *SAZ*. Left: the task-specific results, each averaged over five repetitions. Right: the unweighted task mean. The classical methods use LPQ; the CNN uses pixels. Inferential contrasts concern the fixed predictions and this selected representation.

bility, a reliability-weighted mean, and majority voting. Sigmoid calibration uses the three inner participant folds with ensemble averaging. Reliability is derived from inner-validation macro F1, with weight $\max(F_{\text{inner}} - 0.5, 0.01)$. We do not select a subset of tasks according to outer-test performance. A missing spiral is omitted from that participant's denominator.

The probability mean, specified before the fusion-rule comparison, reaches macro F1 0.6341, with interval [0.5417, 0.7189]. Reliability weighting reaches 0.6447 and majority voting 0.6488. Majority minus probability mean is 0.0147, with interval [−0.0317, 0.0618] and adjusted $p = 1.0000$; none of the three rule comparisons survives correction.

We use majority voting as a common descriptive participant endpoint for all four classifiers. If $\mathcal{T}_j$ is the set of tasks available for participant $j$, its score in repetition $r$ is

$$s_{j,r} = \frac{1}{|\mathcal{T}_j|} \sum_{t \in \mathcal{T}_j} \widehat{y}_{j,t,r}, \qquad \widehat{y}_{j,r} = \mathbf{1}[s_{j,r} \geq 0.5]. \quad (8)$$

Tied votes are assigned to PD by this fixed threshold. We compute the participant-level metric separately for each repetition and average the five scores. We do not average probabilities across repetitions to create another ensemble.

Figure 6 places logistic regression at 0.6498 and SVM at 0.6488 after fusion, both with mean accuracy 0.6507. Random Forest reaches macro F1 0.6259 and the CNN 0.5544. Logistic regression's macro F1 exceeds SVM by only 0.0010, with interval [−0.0398, 0.0403] and adjusted $p = 1.0000$. No pairwise fusion contrast remains significant after Holm correction. Failure to detect a difference is not a formal equivalence result.

The fused SVM has PD sensitivity 0.7135 and H specificity 0.5895. Logistic regression has sensitivity 0.7081 and specificity 0.5947. Their error rates are not symmetric despite near-balanced class counts. The PD decision on tied votes contributes to the interpretation of this operating point; no alternative threshold was chosen from the outer labels. The vote fraction gives an SVM AUC of 0.6808 and a logistic-regression AUC of 0.6658, even though logistic regression ranks slightly higher by thresholded macro F1.

These participant-level scores and the earlier task means summarize different predictions; their numerical difference is not tested as a paired improvement.

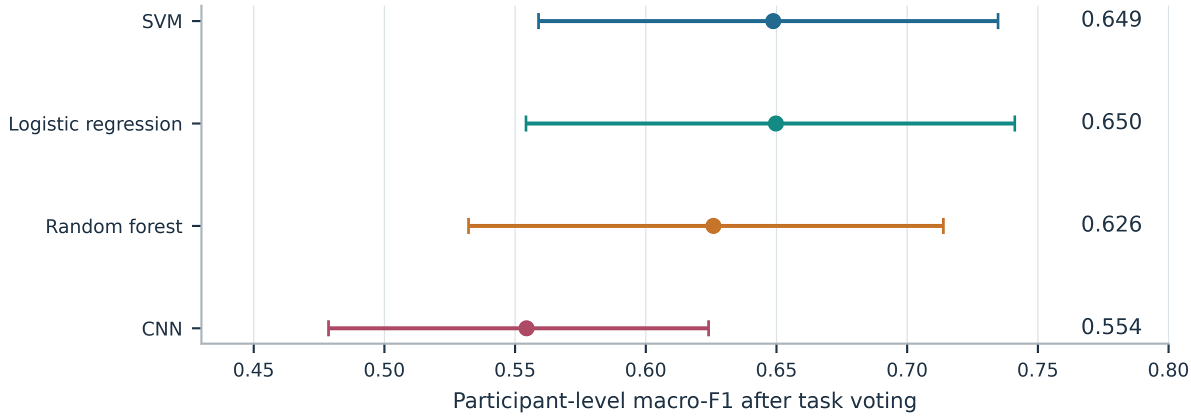


Figure 6: One prediction per participant obtained by voting over available tasks. Points average macro F1 over the five repetitions; intervals are conditional 95% participant-bootstrap intervals. These scores have a different prediction unit from the task means. None of the six paired fusion comparisons remains significant after Holm adjustment.

## 7 WHAT DOES THE CNN USE?

The preceding classification scores leave open how the CNN uses the encoded images. We examine whether its predictions depend on the handwriting region or on image placement and background, then compare spatial explanations and responses to perturbations.

### 7.1 Explanations of Held-Out Predictions

We apply XAI after freezing all outer checkpoints. Each image is explained by the exact checkpoint that produced its held-out prediction, rather than by a model fitted later to the entire cohort.

Grad-CAM weights convolutional activation maps with gradients of a target score (Selvaraju et al., 2017). We target the originally predicted class and use the last ReLU before the fourth pooling layer. At this point the spatial map is $16 \times 16$ for a $128 \times 128$ input. For activation map $A^k$ and predicted-class logit $g_c$, the attribution is

$$L_c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad \alpha_k^c = \frac{1}{UV}\sum_{u,v}\frac{\partial g_c}{\partial A_{uv}^k}. \quad (9)$$

For the single output logit $g$, the PD target is $g_c = g$ and the H target is $g_c = -g$. The positive map is interpolated to input size and divided by its maximum when nonzero. Heatmap colours express attribution magnitude, independently of the input's semantic channels.

We calculate Grad-CAM for all 2,985 held-out predictions. A separate occlusion analysis replaces each non-overlapping $16 \times 16$ input patch with white and measures the decrease in confidence for the original predicted class. Negative changes are clipped to zero for the spatial occlusion map. This analysis covers all 597 predictions in the first repetition.

Figure 7 illustrates correct and incorrect T4 predictions from both diagnosis classes, selected by the rule in its caption before inspecting the explanations. The aggregate analyses cover every task.

### 7.2 Spatial Concentration and Perturbation

We define foreground as pixels with at least one channel below 250, matching the preprocessing. Foreground enrichment is the fraction of attribution inside that mask divided by the fraction of image area occupied by the mask. A uniform nonzero attribution map has enrichment one.

The participant-averaged Grad-CAM enrichment is 1.803, with interval [1.723, 1.884], indicating concentration around the rendered writing. There are 174 all-zero Grad-CAM maps, 5.83% of the predictions. Enrichment is undefined for those maps, which are omitted before aggregation.

To test sensitivity to the highlighted pixels (Figure 8), we whiten the 10% of input pixels with the greatest Grad-CAM values. Confidence in the original class falls by 0.0489 on average, with interval [0.0427, 0.0553]. An equal-area random deletion gives a drop of 0.0033 [0.0016, 0.0050]. The paired advantage of the directed deletion is 0.0456 [0.0395, 0.0518]. Before bootstrapping, repeated predictions are averaged within each participant and task, and tasks are averaged within participant.

Deletion summaries retain all 2,985 predictions, including the 174 all-zero maps. For those maps, the

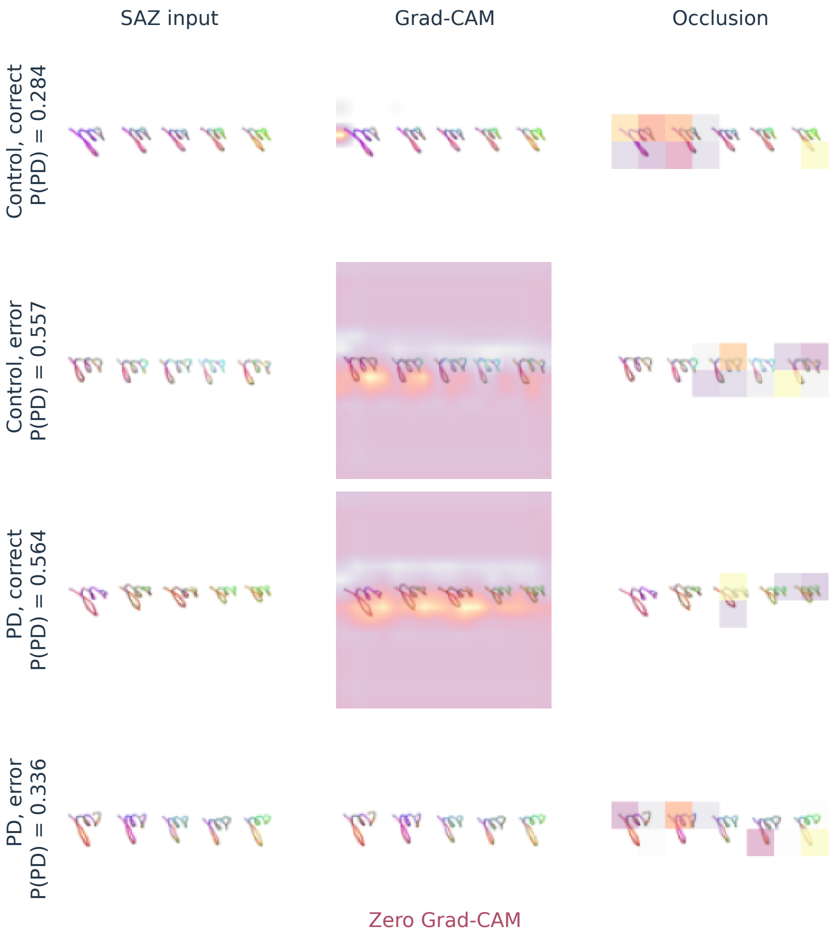


Figure 7: Task-4 explanations for correct and incorrect predictions from both classes. The four cases come from the first repetition, selected at median predicted-class confidence within each diagnosis/outcome stratum. Maps target the original predicted class and use its outer-test checkpoint. Opacity follows per-map attribution. The all-zero Grad-CAM in the PD error is retained. The CNN output is not a calibrated clinical probability.

selected pixels follow the ranking routine's tie handling and carry no attribution-based preference.

The random control matches the number of replaced pixels but not their overlap with foreground or spatial shape. Whitening can also create inputs outside the training distribution. The deletion contrast therefore measures sensitivity under these perturbations and does not fully establish attribution faithfulness.

The two attribution methods show little spatial agreement. Median pixelwise Spearman correlation between Grad-CAM and occlusion is $-0.0121$ across 540 nonconstant pairs; the remaining 57 correlations are undefined. Concentration on handwriting therefore does not imply agreement about which regions influence the prediction.

We also reinitialize all network weights and compare Grad-CAM for one deterministically chosen held-out image per checkpoint. The 200 checks cover only 11 distinct participants because the selection rule repeatedly chooses the first identifier in each test partition. The median trained–random correlation is 0.0276 among 160 nonconstant pairs. Low agreement is consistent with dependence on learned weights, but the limited participant coverage and excluded constant maps constrain this check.

### 7.3 Sensitivity to the Encoded Channels

Replacing a whole colour channel with white tests the CNN's sensitivity to that input component. Across the held-out predictions, the mean confidence drops are 0.0429 for speed, 0.0482 for altitude and 0.0427 for azimuth. Their intervals are [0.0343, 0.0512], [0.0400, 0.0568] and [0.0360, 0.0494], respectively.

These similar global values conceal task varia-

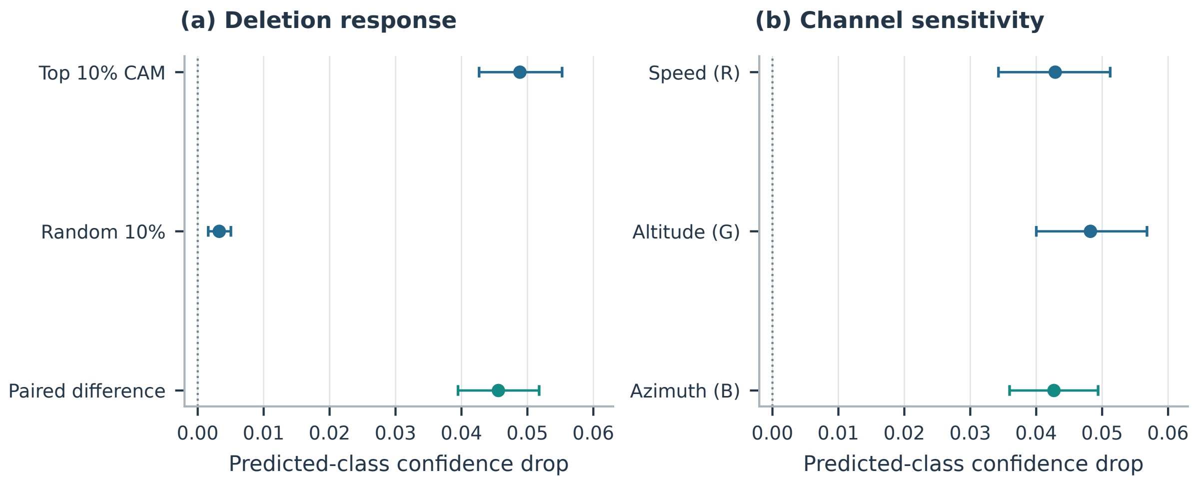


Figure 8: Confidence changes for the originally predicted class. Left: deleting the top 10% Grad-CAM pixels versus equal-area random deletion. Right: whitening each semantic channel. Intervals resample participants after averaging repeats and tasks. These perturbations measure sensitivity of the frozen CNN; the random control does not match foreground overlap, and channel removal also changes colour contrast.

tion. Speed has the largest average effect on the spiral, while its effect on T6 is slightly negative. Removing a channel alters colour contrast as well as its coded measurement. The perturbation therefore cannot rank the underlying physiological signals or settle the speed-versus-pressure choice in the earlier SVM experiment.

These checks show sensitivity to writing regions and to all three input channels. They do not identify a disease-specific image feature.

# 8 DISCUSSION

The observed encoding gains vary across tasks, while LPQ-based models outperform the tested CNN on the selected encoding. We interpret these findings in terms of the information retained by the representation, compare them with earlier PaHaW studies, and examine the limits of the evaluation and attribution analyses.

## 8.1 What the Incremental Comparison Establishes

*SAZ* improves observed macro F1 on four tasks, with an increase of 0.0802 on the spiral. Gains and losses across tasks produce a smaller mean increase of 0.0109, whose interval includes zero. The result is a task-dependent response to the tested colour encoding and LPQ descriptor, with insufficient evidence for overall superiority.

The renderer removes absolute size and signal levels, LPQ discards the location of each phase code, and the image omits in-air movement. These choices may discard useful information alongside nuisance variation. The present experiment cannot separate their effects.

Task dependence remains substantial: T4 is strongest for static SVM and *SAZ* Random Forest, whereas the spiral is strongest for *SAZ* SVM. The task average captures this variation, while fusion asks whether combining their predictions is useful for a participant-level decision.

## 8.2 Comparison with Handwriting Studies

Casademunt González (Casademunt González, 2023) evaluates CNNs on PaHaW crops of size 111 for T2, T3 and T4. The code separates 62 training and 13 test participants before producing crops. Its F1 concerns the positive PD class, rather than macro F1.

Table 3 compares reported accuracy with the task set identified. Casademunt's T2/T3/T4 accuracies exceed our *SAZ* SVM values of 51.20/58.13/64.00% and CNN values of 55.20/52.00/53.60%. This is not a paired comparison: the splits, number of training crops and selection procedures differ, and the thesis compares crop sizes using the designated test group.

Drotár et al. report 81.3% accuracy for kinematic and pressure features from T2–T8 (Drotár et al., 2016). Their descriptors can retain physical information removed by our normalization. They repeat tenfold validation ten times and screen features statisti-

Table 3: Contextual comparison on PaHaW. All values in this table are accuracy, not macro F1. Published results are transcribed from the cited works; none was rerun under our partitions. Different task sets and selection procedures prevent a direct superiority test.

| Study | Input and classifier | Evaluation scope | Accuracy (%) |
|---|---|---|---|
| Casademunt González (2023) | CNN, handwriting crops | T2 / T3 / T4; 62 train, 13 test | 69.23 / 61.54 / 69.23 |
| Drotár et al. (2016) | Kinematics + pressure, SVM | T2–T8; repeated ten-fold CV | 81.30 |
| Diaz et al. (2019) | Enhanced images and task ensemble | Ten-fold CV; five selected tasks | 86.67 |
| Present experiment | *SAZ* + LPQ + SVM | T1–T8; $5 \times 5$ participant CV; majority fusion | 65.07 |
| Present experiment | *SAZ* + LPQ + logistic regression | T1–T8; $5 \times 5$ participant CV; majority fusion | 65.07 |

cally; the description does not fully establish whether screening is nested in each split.

Diaz et al. report 86.67% accuracy for their enhanced representation and ensemble, versus 72.50% for its static counterpart (Diaz et al., 2019). The enhancement includes in-air movement, and their pipeline uses pretrained CNN features and five selected tasks. It preserves and models different information from our contact-only colour images with LPQ. These protocol differences preclude a controlled cross-paper ranking.

Our participant-level accuracy is lower than the published combined-task results of Drotár and Diaz. The present study does not determine how much of this gap arises from representation, classifier or evaluation choices.

## 8.3 Limits of Complexity and Explanation

Each task provides approximately 60 outer-training participants for learning filters from scratch; augmented images remain correlated views of those records. The observed LPQ advantage is conditional on this cohort and the selected representation. Differences in input resolution and model-selection procedures also prevent attributing it solely to network architecture.

The near-zero Grad-CAM–occlusion correlation limits the consistency of spatial interpretation for this CNN. The maps alone do not establish disease-specific features or explain the classifier's lower performance.

Independent-cohort validation is needed to estimate generalization beyond this development process, including the effects of age imbalance and acquisition differences. A new random split of PaHaW would still reuse the cohort involved in design decisions. Further evaluation should select representations within training partitions or freeze them before testing on a new cohort.

## 8.4 Reproducibility

The accompanying experimental source package supplies the rendering, feature extraction, model fitting, fusion and XAI code, together with the eight stage configurations and pinned dependencies. Reproduction requires access to the original PaHaW recordings, which are not redistributed in this package. Its README gives the execution order and distinguishes model reruns from figure generation using the archived results.

Our internal experimental archive retains participant assignments, rendering diagnostics, inner searches, outer predictions and CNN checkpoints. XAI checks reproduce checkpoint probabilities before explanation, and checksums link figure inputs and tables to their source files.

# 9 CONCLUSION

We introduced a formulation of pseudodynamic handwriting minimap signatures that maps normalized pen signals onto contact-only trajectories. Its incremental evaluation shows observed gains on four of eight tasks for speed, altitude and azimuth, with the largest increase on the spiral. The task mean rises from 0.5643 to 0.5752, while the paired analysis leaves the overall advantage over static uncertain.

On this SVM-selected encoding, LPQ-based classifiers outperform the tested CNN in the conditional task-mean comparison. Majority voting yields participant-level macro F1 near 0.65 for logistic regression and SVM. Grad-CAM concentrates on writing, but its poor agreement with occlusion limits the consistency of spatial interpretation. An independent-cohort test, with the encoding fixed in advance, is needed to assess whether these task-dependent gains generalize.

# REFERENCES


Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

Casademunt González, A. (2023). Análisis automático de alteraciones en la escritura manuscrita debidas al párkinson. Undergraduate thesis, Universidad Rey Juan Carlos. Academic year 2022/2023.

Cilia, N. D., D'Alessandro, T., De Stefano, C., Fontanella, F., and Molinara, M. (2021). From online handwriting to synthetic images for Alzheimer's disease detection using a deep transfer learning approach. *IEEE Journal of Biomedical and Health Informatics*, 25(12):4243–4254.

Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3):273–297.

Diaz, M., Ferrer, M. A., Impedovo, D., Pirlo, G., and Vessio, G. (2019). Dynamically enhanced static handwriting representation for Parkinson's disease detection. *Pattern Recognition Letters*, 128:204–210.

Drotár, P., Mekyska, J., Rektorová, I., Masarová, L., Smékal, Z., and Faundez-Zanuy, M. (2016). Evaluation of handwriting kinematics and pressure for differential diagnosis of Parkinson's disease. *Artificial Intelligence in Medicine*, 67:39–46.

Gebara Jr., M., Wiese, I. S., and Costa, Y. M. G. (2025). Software feature detection using source code minimaps as visual signatures. In *32nd International Conference on Systems, Signals and Image Processing (IWSSIP)*. IEEE.

Impedovo, D. and Pirlo, G. (2019). Dynamic handwriting analysis for the assessment of neurodegenerative diseases: A pattern recognition perspective. *IEEE Reviews in Biomedical Engineering*, 12:209–220.

LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324.

Moetesum, M., Siddiqi, I., Vincent, N., and Cloppet, F. (2019). Assessing visual attributes of handwriting for prediction of neurological disorders — a case study on Parkinson's disease. *Pattern Recognition Letters*, 121:19–27.

Ojansivu, V. and Heikkilä, J. (2008). Blur insensitive texture classification using local phase quantization. In *Image and Signal Processing (ICISP)*, volume 5099 of *Lecture Notes in Computer Science*, pages 236–243. Springer.

Pereira, C. R., Weber, S. A. T., Hook, C., Rosa, G. H., and Papa, J. P. (2016). Deep learning-aided Parkinson's disease diagnosis from handwritten dynamics. In *29th SIBGRAPI Conference on Graphics, Patterns and Images*, pages 340–346.

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision (ICCV)*, pages 618–626. IEEE.